Fundamentals of AI and ML

Project Report

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Abstract

In our increasingly globalized world, language barriers remain a significant challenge for effective communication. This project aims to bridge that gap by developing a Voice-to-Voice Translation system. By leveraging Python libraries such as SpeechRecognition, gTTS, and Translator, the application captures spoken input in one language, translates it into a target language, and plays back the translated text as audio.

Introduction

The objective of this project is to create an accessible tool for travelers, students, and professionals to communicate across linguistic boundaries. The system utilizes Google's speech recognition engine and the Google Text-to-Speech (gTTS) API to ensure high accuracy and natural-sounding output.

Software Requirements

 Operating System: Windows/macOS/Linux

 Language: Python 3.x

 Libraries: * SpeechRecognition: For capturing and processing voice input.

- gTTS (Google Text-to-Speech): For converting text to audio.
- translate: For language conversion.
- playsound: For outputting the translated audio.

Hardware Requirements

 High-quality Microphone for input.

 Speakers or Headphones for output.

 Active Internet Connection (required for Google API calls

System Flowchart

 Audio Input: User speaks via the microphone.

 Speech Recognition: The audio is converted to a string using Google's recognizer.

 Language Selection: The user specifies the target language.

 Translation: The text is sent to the translation engine.

 Audio Synthesis: The translated text is converted into an MP3 file.

 Playback: The system plays the translated audio and deletes the temporary file.

Language Dictionary

The system uses a comprehensive tuple containing language names and their corresponding ISO-639-1 codes (e.g., 'hindi': 'hi'). This allows the system to map user verbal input to the specific codes required by the APIs.

Core Functions

- takecommand(): Handles microphone energy thresholds and listens for audio. It handles exceptions where speech might be unintelligible.
- destination_language(): Prompts the user for the target language and validates it against the supported dictionary.
- **The Translation Loop:** Combines the translator and gTTS to synthesize the final output.

Code Logic Highlights

The script uses a while loop to ensure the user provides a valid language before proceeding. It also uses temporary file management (os.remove) to ensure the local storage isn't cluttered with audio files after each translation.